INSEOK LEE

Graduate Researcher
Hydrogen & Gas Systems | Process Simulation & AI Engineering

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Seoul, South Korea
Korean (Native) | English (Advanced)

PROFILE

Chemical engineering graduate researcher specialising in hydrogen gas generation, purification, and supply systems built to meet high pressure gas safety codes. Designed, built, and operated a bench scale hydrogen production and purification system integrated with a PEM fuel cell, reducing CO from ~10,000 ppm to below 1 ppm and validating 48 h continuous operation. Handled hazardous area and explosion protection compliance for the purification unit, and builds AI assisted tools such as an Aspen Plus automation pipeline that cut ~26 h of manual work to under 1 h.

EDUCATION

Korea University — Seoul, South Korea
M.S. in Chemical and Biological Engineering — Mar 2025 – Feb 2027 (expected)

- GPA 4.31 / 4.5 · Process Systems Engineering Lab (Advisor: Prof. Wangyun Won)
- Thesis: Optimal thermal integration of combined-cycle power plants and LOHC hydrogen production systems

Konkuk University — Seoul, South Korea
B.S. in Chemical Engineering — Mar 2018 – Feb 2025

- GPA 3.67 / 4.5 (Major 3.68)

RESEARCH & WORK EXPERIENCE

Korea Institute of Science and Technology (KIST) — Seoul, South Korea
Graduate Student Researcher — Center for Hydrogen and Fuel Cells — Mar 2025 – Present

Hydrogen Generation & Purification System Engineering

- Designed, fabricated, and operated a bench-scale $NaBH_4$ hydrolysis, CO purification, and PEMFC power system across all units (reactor, gas–liquid separator, purifier, fuel cell), including thermal management of the strongly exothermic hydrolysis (approx. −210 kJ/mol $NaBH_4$).
- Owned the CO purification unit: a single thermochemical unit (molecular sieve, $Ru/TiO_2/Al_2O_3$ catalyst, 13X) that converts CO to condensable C_{3+} hydrocarbons via a Fischer–Tropsch-type reaction instead of PROX, adsorption, or methanation, driven by hydrolysis waste heat without auxiliary reagents (120–160 °C, 20–40 barg); lowered CO from ~10,000 ppm to <1 ppm and validated with 48 h continuous integrated operation.
- Operated an IV / EIS evaluation protocol that detects CO poisoning of the PEMFC catalyst before output degradation appears.
- Conducted as joint research with the Hyundai Motor Fuel Cell Development Center; published in the International Journal of Hydrogen Energy as co-first author (Aug 2026).

Certification & Safety Compliance — Portable Hydrogen Power System

- Reviewed hazardous-area classification (KGS GC201) and explosion-proof equipment requirements (KGS GC205, Ex d IIC T6, KCs certification) under the Korean High-Pressure Gas Safety Control Act, and translated them into component-level specifications for the certification and safety review of a portable hydrogen power system.
- Scaled the system to 1 kg H_2 production and verified purification performance with a 2 kW-class PEMFC.
- Responsible for requirement definition, performance-verification data, and review documentation of the CO purification unit in a three-party structure (prime contractor – engineering firm – KIST).
- Self-taught Autodesk Inventor (40+ practice drawings) and produced the purifier 3D drawings in-house, cutting drawing-revision turnaround from 3–4 days (outsourced) to same-day.

Process Simulation & Thermal Integration

- M.S. thesis: Aspen Plus modelling of heat integration between a combined-cycle power plant and LOHC dehydrogenation; showed that the marginal power penalty is governed by the HRSG pinch structure rather than extraction temperature, and derived demand-dependent extraction-line and split-ratio selection rules.
- Built a Python ODE thermal-simulation framework, validated against bench temperature data, to compare 5 heat-recovery configurations; showed that driving the CO purifier with hydrolysis waste heat improves cooling and purification simultaneously.

Automation & AI-Driven Engineering Tools

- Aspen Plus automation pipeline (Python COM, Mar 2026–): DOE generation, parallel execution, fault isolation, and result aggregation; completed 1,575 simulation cases unattended in under 1 h vs. ~26 h of manual GUI work.
- Interactive web dashboard (Chart.js / SheetJS) for exploring simulation results with advisor and collaborators; removed 2–3 h of per-meeting data compilation and enabled in-meeting review.
- Lab knowledge-base RAG system (Jul 2026–): indexed lab notebooks and papers with natural-language search, deployed to the lab via MCP; retrieval of past experimental conditions reduced from tens of minutes to seconds.
- Claude Code orchestration of 11 sub-agents and 30+ skills with a cross-LLM review loop, keeping output quality consistent as AI models change.

Key Funded Projects (participating, 2026):

- Integrated Source Technology for Green Hydrogen Production and Liquid-Phase Storage — Future Original Clean Hydrogen Convergence Technology Program, KIST-led (Jan – Dec 2026)
- Development of a Hydrogen Fuel Cell Power Source for Wheeled Armoured Vehicles — Materials & Components Technology Development Program (Jan – Dec 2026)
- Fabrication and Commercialisation Demonstration of a 45 kW Direct-Ammonia-Fed Fuel Cell Power Generation System — Ministry of SMEs and Startups (Jan – Dec 2026)

Research Intern — Center for Hydrogen and Fuel Cells Jun 2024 – Aug 2024

- Set up a PEMFC single-cell test station: cell assembly, H_2 / air supply line configuration, leak checks, and IV / EIS evaluation under varied hydrogen utilisation.
- Ran daily operation and inspection of the test facility, including hydrogen supply piping, mass flow control, and safety interlocks; automated IV / EIS data processing and plotting in Python.

PUBLICATIONS

Selective CO-to-Hydrocarbon Conversion in Formic Acid–Promoted Borohydride Hydrolysis for PEMFC-Grade Hydrogen Production
Co-first Author | International Journal of Hydrogen Energy, Vol. 270, 157136 (Aug 2026) | doi:10.1016/j.ijhydene.2026.157136

Optimal Thermal Integration of CCPP and LOHC Hydrogen Production Systems
Co-first Author | Manuscript in preparation (M.S. thesis work)

Thermal Integration of a CO Purifier with $NaBH_4$ Hydrolysis Waste Heat: Comparison of Heat-Recovery Configurations
Manuscript in preparation

CONFERENCE PRESENTATIONS

- **KIMST 2026 Annual Conference** (Domestic, Oral — Special Session “Future of K-Defense”) Selective CO-to-Hydrocarbon Conversion in Formic Acid-Promoted Borohydride Hydrolysis for High-Purity Hydrogen Production
- **KHNES Spring Meeting 2026** (Domestic, Poster) Selective CO-to-Hydrocarbon Conversion in Formic Acid-Promoted Borohydride Hydrolysis for High-Purity Hydrogen Production

AWARDS

- **Encouragement Award** — 2nd Defense Materials & Parts R&D Requirements Competition, KRIT (Jul 2025): standalone H_2 refuelling station using a solid-state hydrogen carrier (team of 3)
- **Prize Winner** — K-3MT (Three Minute Thesis) Competition, Korea University Graduate School (Nov 2025)

TRAINING & CERTIFICATIONS

- Anthropic official courses: Claude 101 · Claude Code 101 · Claude Code in Action · Claude Cowork

SKILLS

Process Simulation	Aspen Plus (Advanced) · Python COM automation · ODE thermal modelling
System Engineering	H_2 generation / purification BoP · PEMFC operation · Hazardous-area & explosion-proof design (KGS GC201 / GC205)
Electrochemical Testing	IV polarisation · EIS · CO-poisoning diagnostics
Optimisation & Data	Surrogate modelling · NSGA-II · Bayesian optimisation · DOE
CAD / Design	Autodesk Inventor (Intermediate)
Programming & AI	Python (Advanced) · JavaScript · R · Claude Code multi-agent orchestration · MCP · RAG
Languages	Korean (Native) · English (TOEIC Speaking AL 160, May 2026)